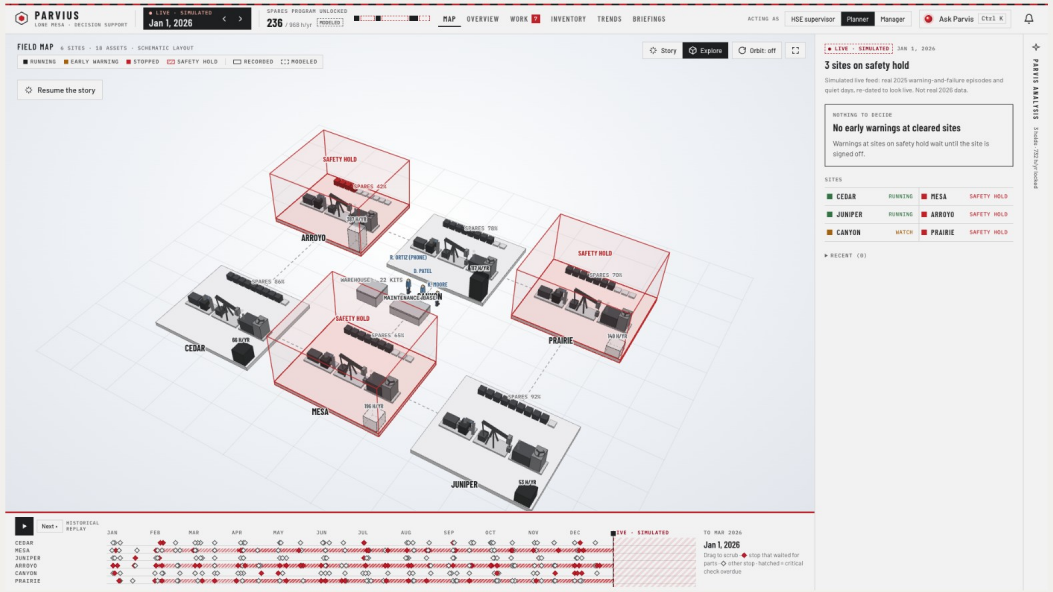
THE PROBLEM

MACHINES WAIT FOR PARTS

24%

of all downtime is spent waiting for a spare part



- 6,599 h lost to breakdowns in 2025
- 24% of it = waiting for a spare part
- Arroyo: 40% of the waiting, only 42% stocked

THE DATA

CLEANED, THEN PROVEN

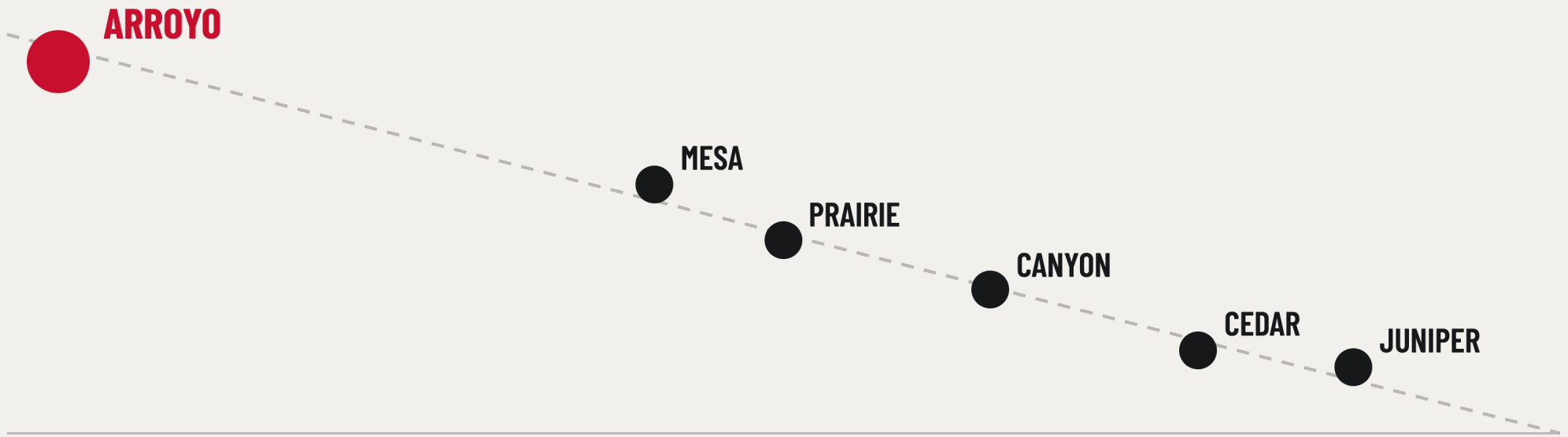
39 / 39

checks pass. Nothing guessed, nothing invented.

- Duplicates removed · IDs and units fixed
- Missing values rebuilt by formula or left blank
- Limits: synthetic · daily readings · 6 sites

THE FINDING

EMPTY SHELVES MEAN WAITING

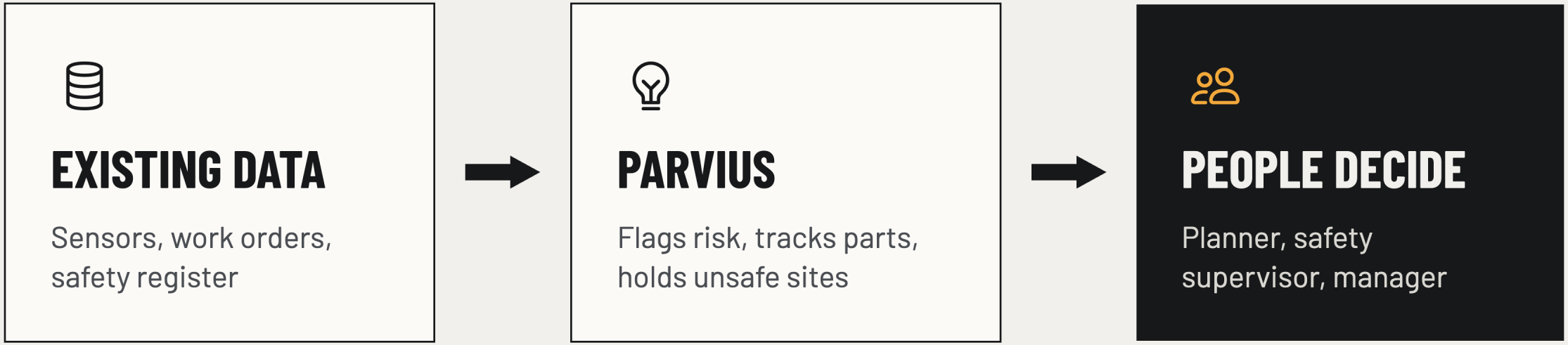


Fuller spare shelf → · higher = more breakdowns waited for parts

- Each dot = one site
- Arroyo 58% empty → 56% waited · Juniper only 10%
- Every wait costs 16–18 h, whatever broke

HOW IT WORKS

DATA IN. PEOPLE DECIDE.



- No new systems: data the company already has
- Flags risk · tracks parts · holds unsafe sites
- No engineer sent without the parts · all logged

THE LOGIC

EVERY NUMBER TRACES BACK

WE SAW

24% waiting

→

CASE SAYS

–60% wait

→

WE DO

Stock spares

→

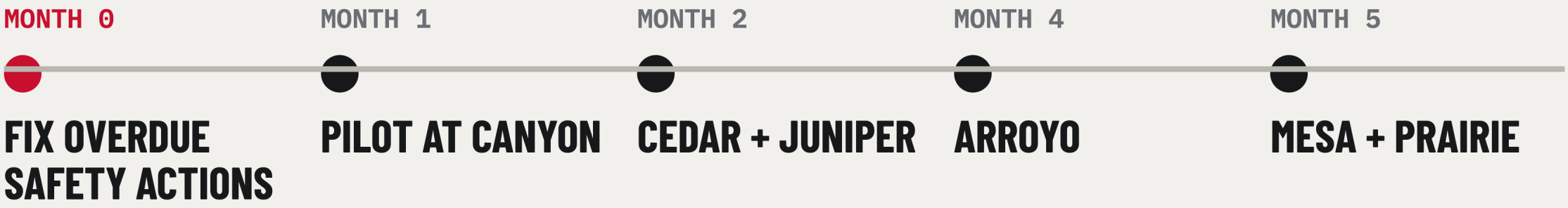
WE GET

968 h / yr

- 60% cut comes from the case's own sheet (35–75%)
- 666 h in year 1, then 968 h every year
- Not claimed: injuries, repair savings, AI hours

THE ROLLOUT

SAFETY FIRST, THEN SITE BY SITE



- Safety supervisors start; planners follow
- Arroyo only after its safety is verified
- Pilot test: Canyon shelf ≥ 90% by month 3

THE COST

ONE PROGRAM, ONE PRICE

$227.5k

to start, then $19.5k a year

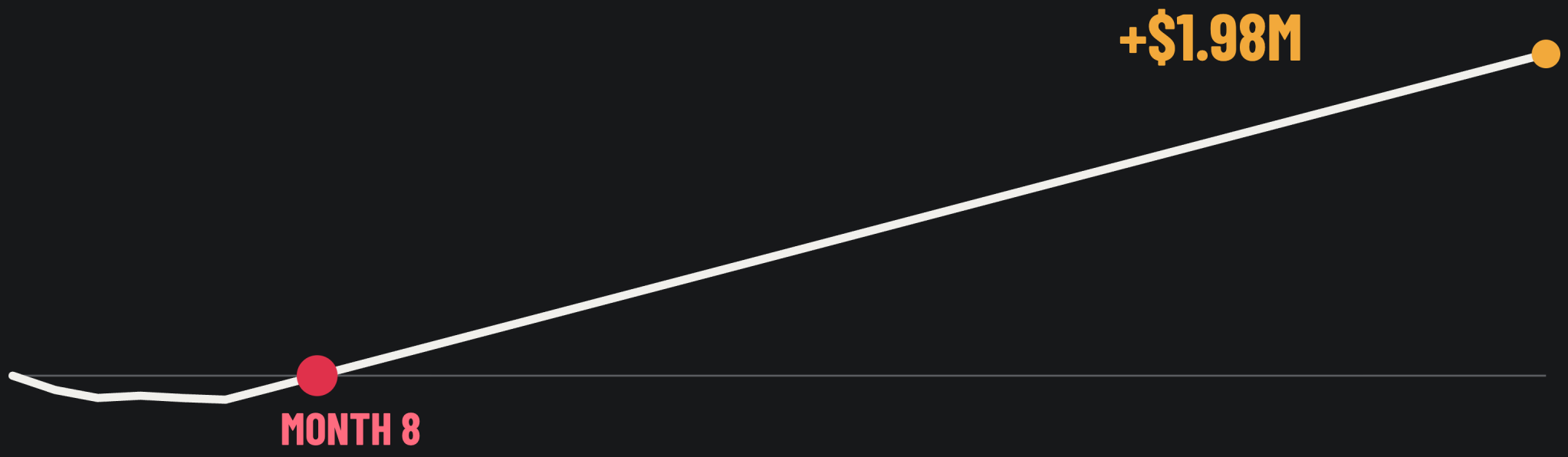
$168k
spare parts, all 6 sites

$59.5k
overdue safety fixes

- $168k spare parts + $59.5k safety fixes
- $33k of the safety work unlocks 3 sites
- Fits staff hours · Arroyo uses 44 of 120

THE RETURN

PAID BACK IN 8 MONTHS



- Small dip while sites are stocked
- ≈ 8× the cost · +$1.98M by year 3
- Worst case still pays back by month 16

THE CASE FOR YES

THREE REASONS

1

BIGGEST FIXABLE DELAY

- Waits hit every site and every failure

2

FAST TO PAY BACK

- 1 month to start · ~8-month payback

3

SAFE BY DESIGN

- Ask: $227.5k for safety sprint + Canyon pilot

SAFETY & RESPONSIBLE USE

SAFETY IS NEVER SKIPPED



NO OVERRIDE

- Open severity-5 action = site can’t start



PEOPLE DECIDE

- Advice only · a person approves and logs it



WORK UNTOUCHED

- Lockout, gas checks and restart unchanged